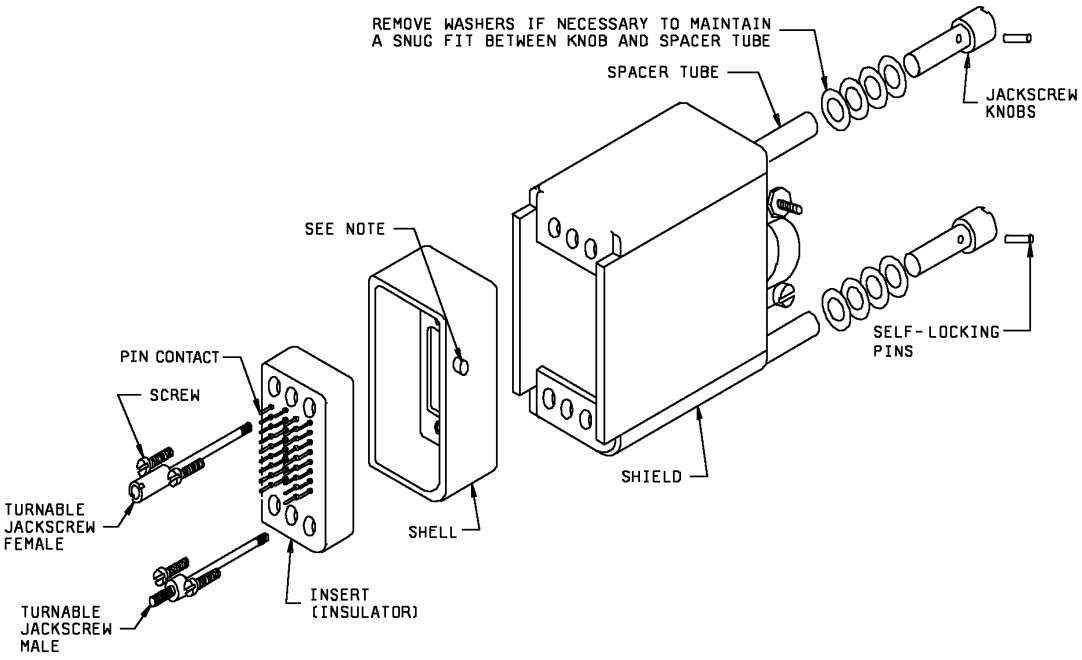
INCH-POUND
MIL-DTL-28748/5E
w/Amendment 1
8 August 2014
SUPERSEDING
MIL-DTL-28748/5E
9 September 2004

DETAIL SPECIFICATION SHEET

CONNECTORS, ELECTRICAL RECTANGULAR,
RACK AND PANEL, SOLDER TYPE PIN CONTACTS, SIZE 20

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-28748.



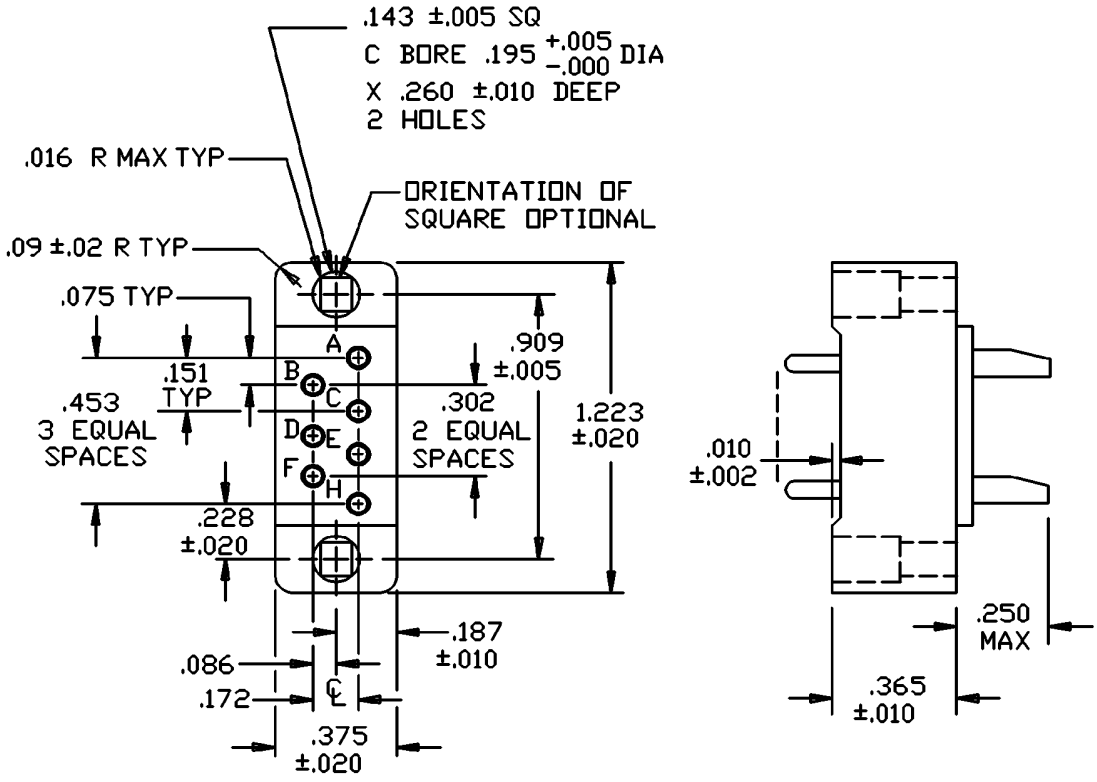
Connector

NOTE: The polarization feature (pin on the plug shell, groove on the receptacle shell) shall be adjacent to the side closest to the number 1 or A contact position.

Configuration A

FIGURE 1. Connector with shield, plug shell, jackscrews, and guidepins.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:12Z
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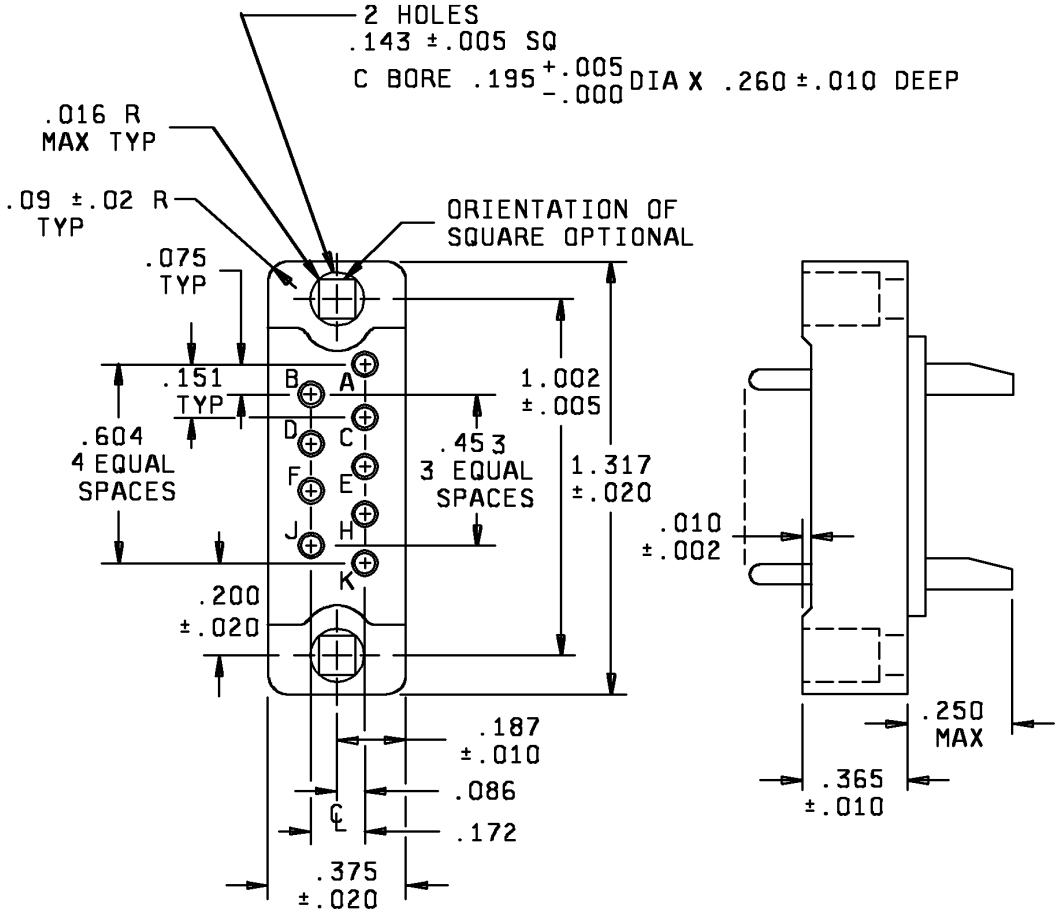
Inches	mm	Inches	mm	Inches	mm
.002	0.05	.09	2.3	.260	6.60
.005	0.13	.143	3.63	.302	7.67
.010	0.25	.151	3.84	.365	9.27
.016	0.41	.172	4.37	.375	9.53
.02	0.5	.187	4.75	.453	11.51
.020	0.51	.195	4.95	.603	15.32
.075	1.91	.228	5.79	.909	23.09
.086	2.18	.250	6.35	1.223	31.06

Insert with 7 contacts

Insert designator A

FIGURE 2. Insert arrangements - Continued.

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Check the source to verify that this is the current version before use.



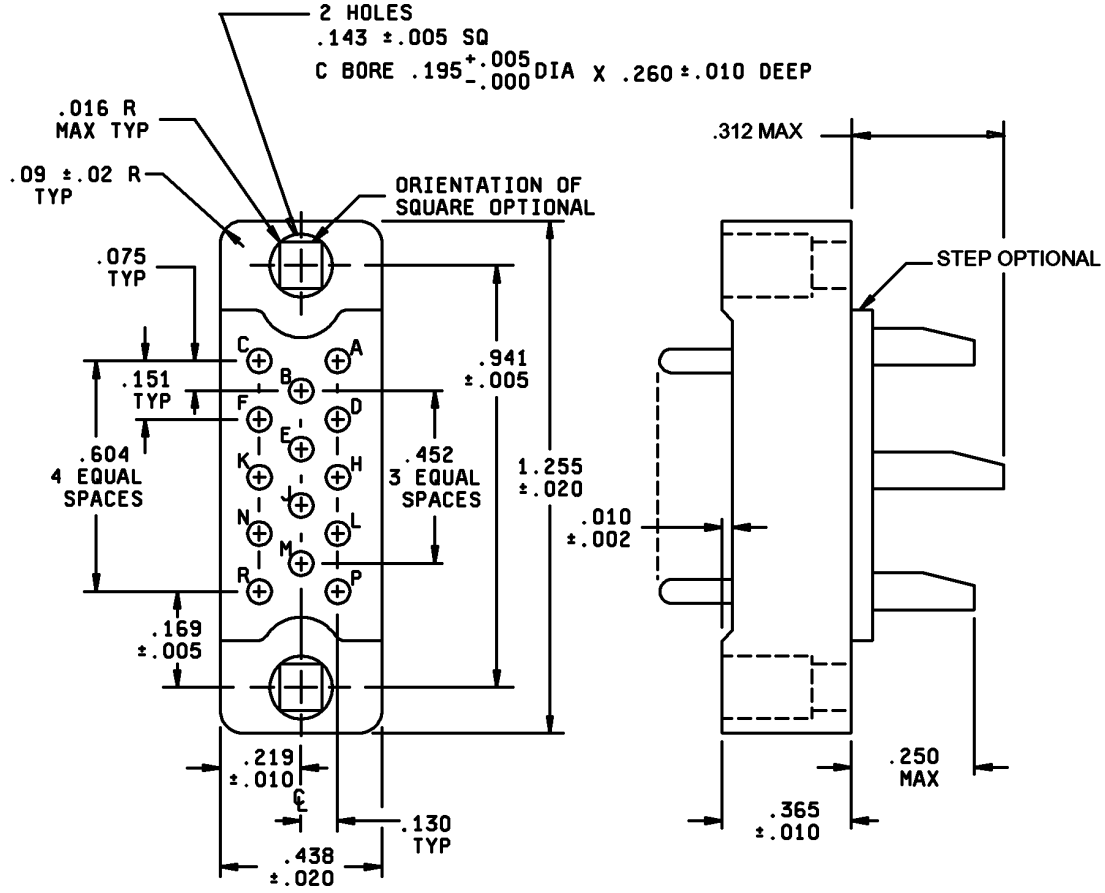
Inches	mm	Inches	mm	Inches	mm
.002	0.05	.09	2.3	.260	6.60
.005	0.13	.143	3.63	.365	9.27
.010	0.25	.151	3.84	.375	9.53
.016	0.41	.172	4.37	.453	11.51
.02	0.5	.187	4.75	.604	15.34
.020	0.51	.195	4.95	1.002	25.45
.075	1.91	.200	5.08	1.317	33.45
.086	2.18	.250	6.35		

Insert with 9 contacts

Insert designator B

FIGURE 2. Insert arrangements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:12Z
Check the source to verify that this is the current version before use.



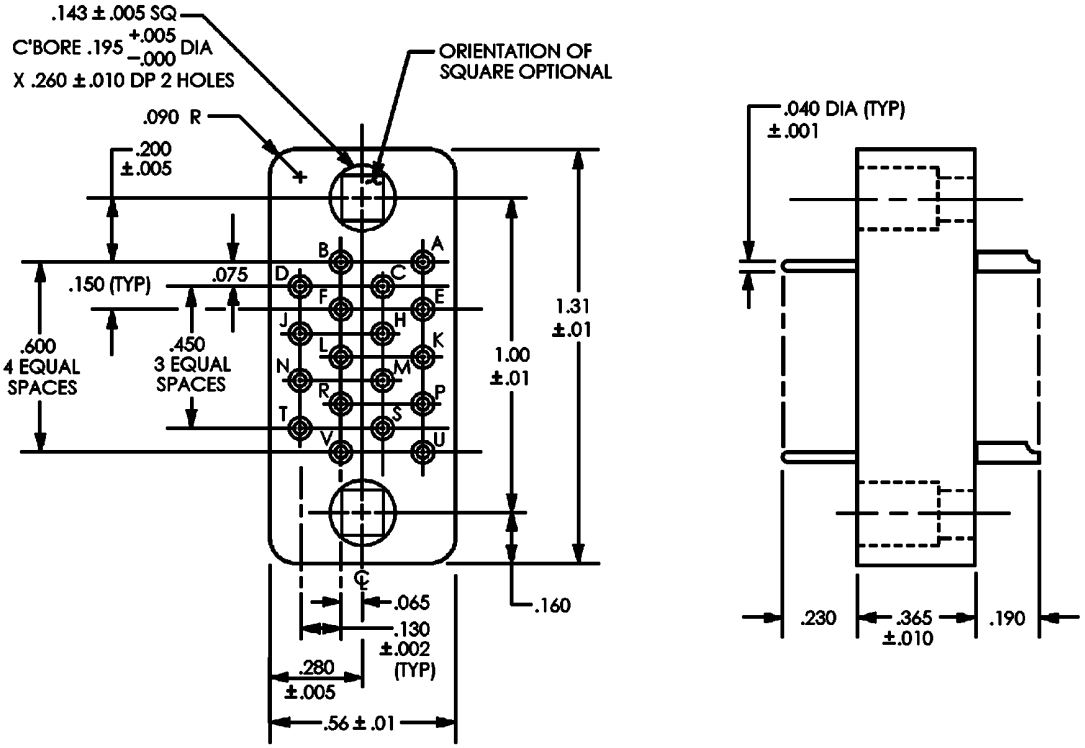
Inches	mm	Inches	mm	Inches	mm
.002	0.05	.130	3.30	.312	7.92
.005	0.13	.143	3.63	.365	9.27
.010	0.25	.151	3.84	.438	11.13
.016	0.41	.169	4.29	.452	11.48
.02	0.5	.195	4.95	.604	15.34
.020	0.51	.219	5.56	.941	23.90
.075	1.91	.250	6.35	1.255	31.88
.09	2.3	.260	6.60		

Insert with 14 contacts

Insert designator C

FIGURE 2. Insert arrangements - Continued.

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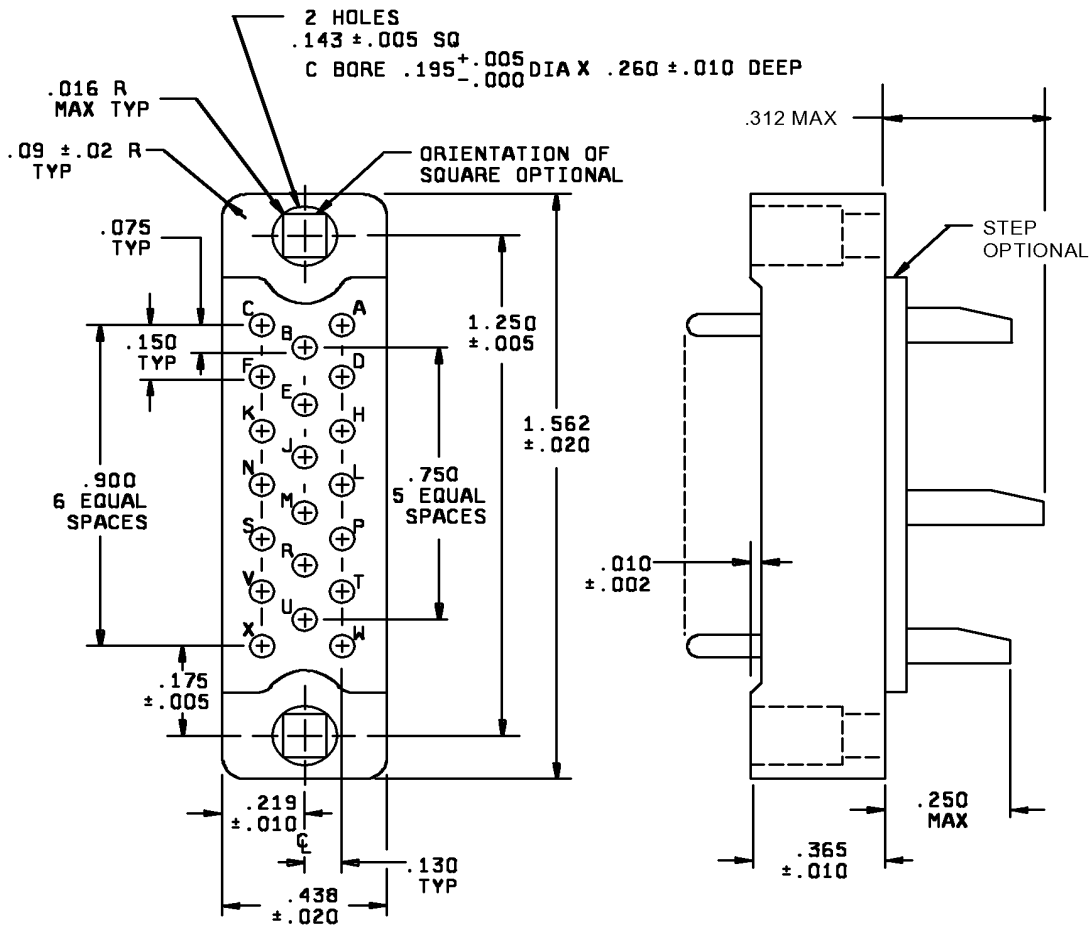
Inches	mm	Inches	mm	Inches	mm
.001	0.03	.090	2.29	.260	6.60
.002	0.05	.130	3.30	.280	7.11
.005	0.13	.143	3.63	.365	9.27
.01	0.3	.160	4.06	.450	11.43
.010	0.25	.190	4.83	.56	14.2
.040	1.02	.195	4.95	.600	15.24
.065	1.65	.200	5.08	1.00	25.40
.075	1.91	.230	5.84	1.31	33.27

Insert with 18 contacts

Insert designator S

FIGURE 2. Insert arrangements - Continued.

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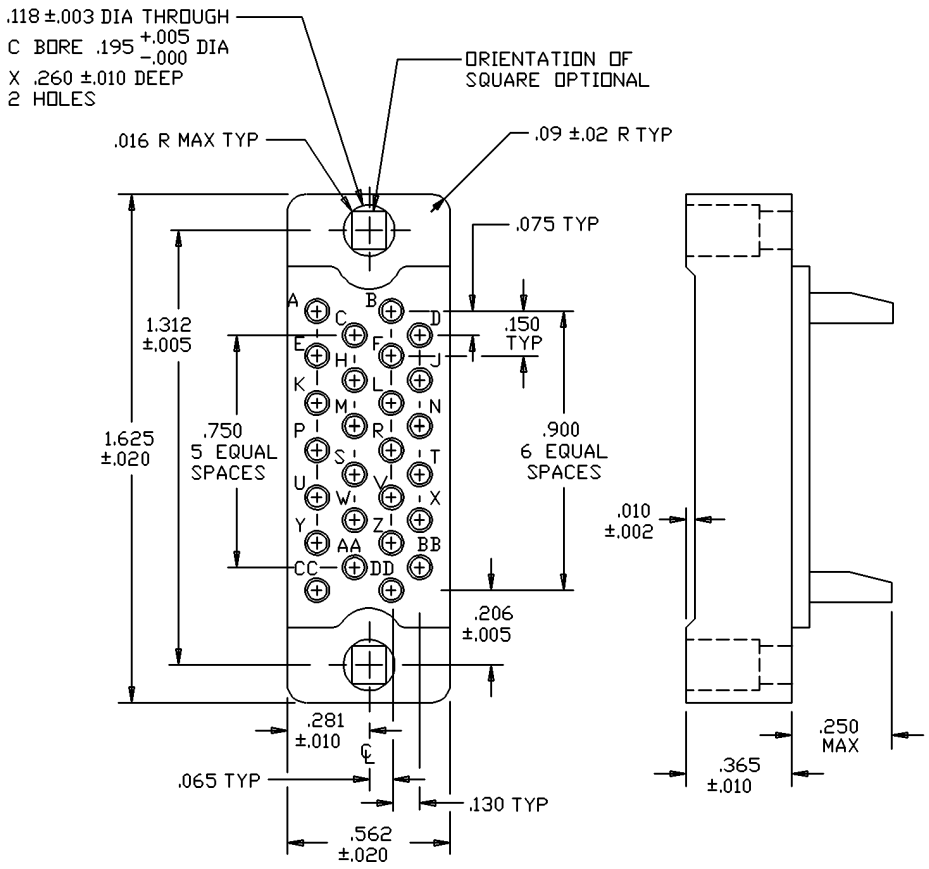
Inches	mm	Inches	mm	Inches	mm
.002	0.05	.09	2.3	.250	6.35
.005	0.13	.130	3.30	.260	6.60
.010	0.25	.143	3.63	.365	9.27
.016	0.41	.150	3.81	.438	11.13
.02	0.5	.175	4.45	.750	19.05
.020	0.51	.195	4.95	.900	22.86
.075	1.91	.219	5.56	1.250	31.75
				1.562	39.67

Insert with 20 contacts

Insert designator D

FIGURE 2. Insert arrangements - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:12Z
Check the source to verify that this is the current version before use.



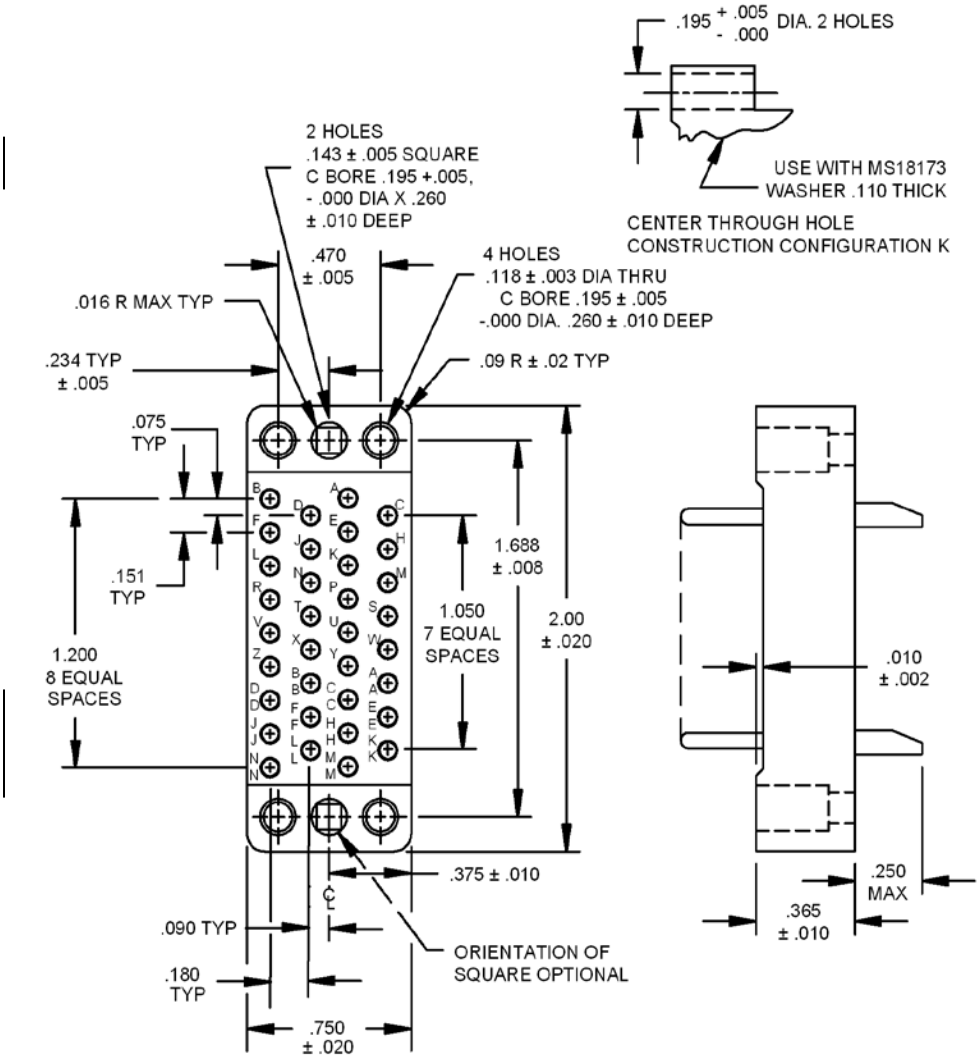
Inches	mm	Inches	mm	Inches	mm	Inches	mm
.002	0.05	.020	0.51	.150	3.81	.365	9.27
.003	0.08	.065	1.65	.195	4.95	.562	14.27
.005	0.13	.075	1.91	.206	5.23	.750	19.05
.010	0.25	.09	2.3	.250	6.35	.900	22.86
.016	0.41	.118	3.00	.260	6.60	1.312	33.32
.02	0.5	.130	3.30	.281	7.14	1.625	41.23

Insert with 26 contacts

Insert designator E

FIGURE 2. Insert arrangements - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:12Z
Check the source to verify that this is the current version before use.



Inches	mm
.002	0.05
.003	0.08
.005	0.13
.008	0.20
.010	0.25
.016	0.41
.02	0.5
.020	0.51
.075	1.91
.09	2.3
.090	2.29
.110	2.79
.118	3.00
.143	3.63
.151	3.84
.180	4.57
.195	4.95
.234	5.94
.244	6.20
.250	6.35
.260	6.60
.365	9.27
.375	9.53
.470	11.94
.750	19.05
1.050	26.67
1.200	30.48
1.688	42.88
2.00	50.8

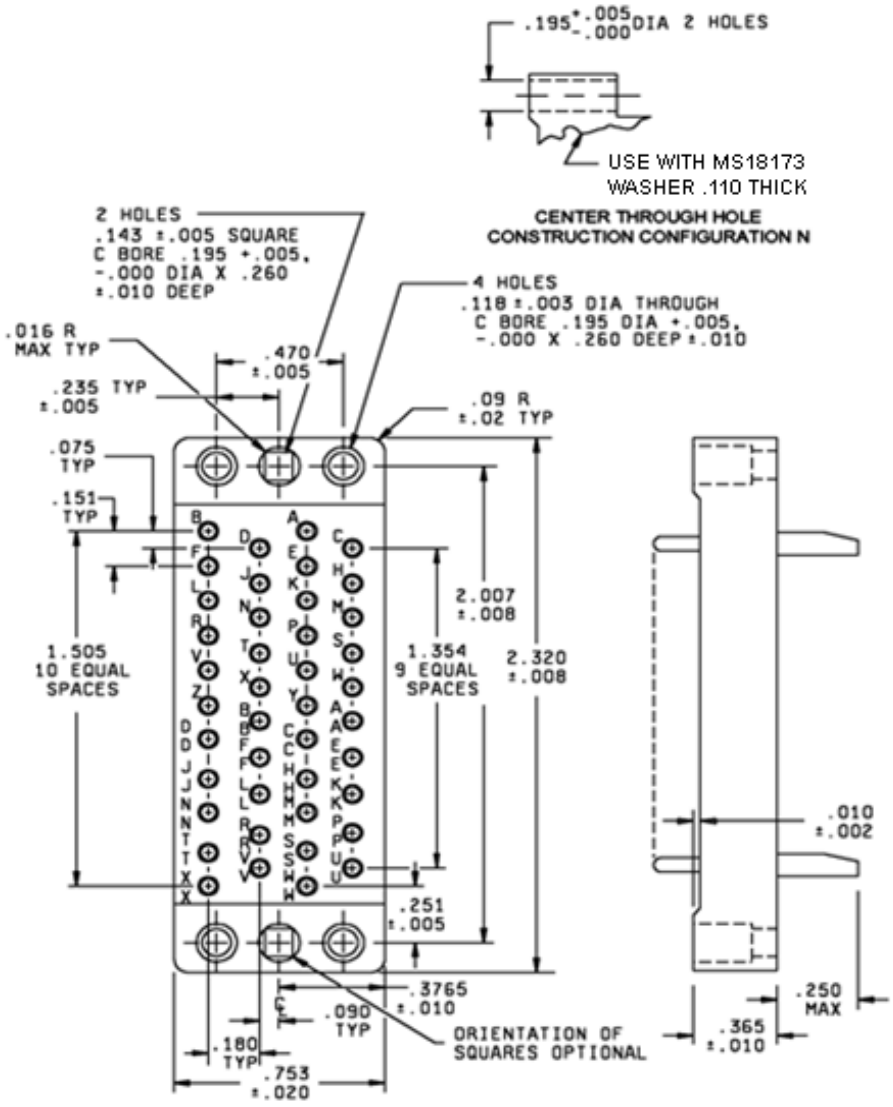
Configuration F - Suitable for all applications, except one-piece jackscrews.
Configuration M - For long jackscrews and shield only (through hole alternate construction).

Insert with 34 contacts

Insert designators F and M

FIGURE 2. Insert arrangements - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:12Z
Check the source to verify that this is the current version before use.



Inches	mm
.002	0.05
.003	0.08
.005	0.13
.008	0.20
.010	0.25
.016	0.41
.02	0.5
.020	0.51
.075	1.91
.09	2.3
.090	2.29
.110	2.79
.118	3.00
.143	3.63
.151	3.84
.180	4.57
.195	4.95
.235	5.97
.250	6.35
.251	6.38
.260	6.60
.365	9.27
.3765	9.563
.470	11.94
.753	19.13
1.354	34.39
1.505	38.23
2.007	50.98
2.320	58.93

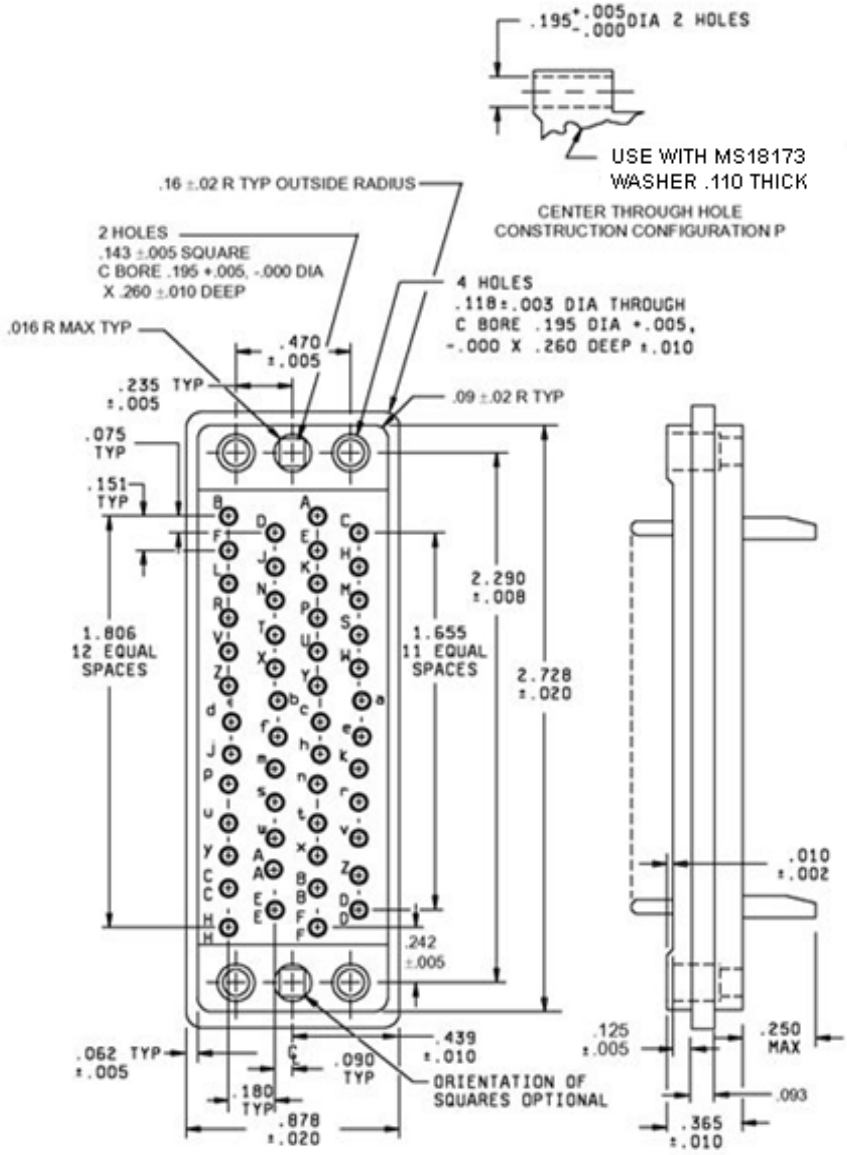
Configuration G - Suitable for all applications, except one-piece jackscrews.
Configuration N - For long jackscrews and shield only (through hole alternate construction).

Insert with 42 contacts

Insert designators G and N

FIGURE 2. Insert arrangements - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:12Z
Check the source to verify that this is the current version before use.



Inches	mm
.002	0.05
.003	0.08
.005	0.13
.008	0.2
.010	0.25
.016	0.41
.02	0.5
.020	0.51
.062	1.57
.075	1.91
.09	2.3
.090	2.29
.093	2.36
.110	2.79
.118	3.00
.125	3.18
.143	3.63
.151	3.84
.156	3.96
.16	4.1
.180	4.57
.195	4.95
.235	5.97
.242	6.15
.250	6.35
.260	6.60
.365	9.27
.439	11.15
.470	11.94
.878	22.30
1.655	42.04
1.806	45.87
2.290	58.17
2.728	69.29

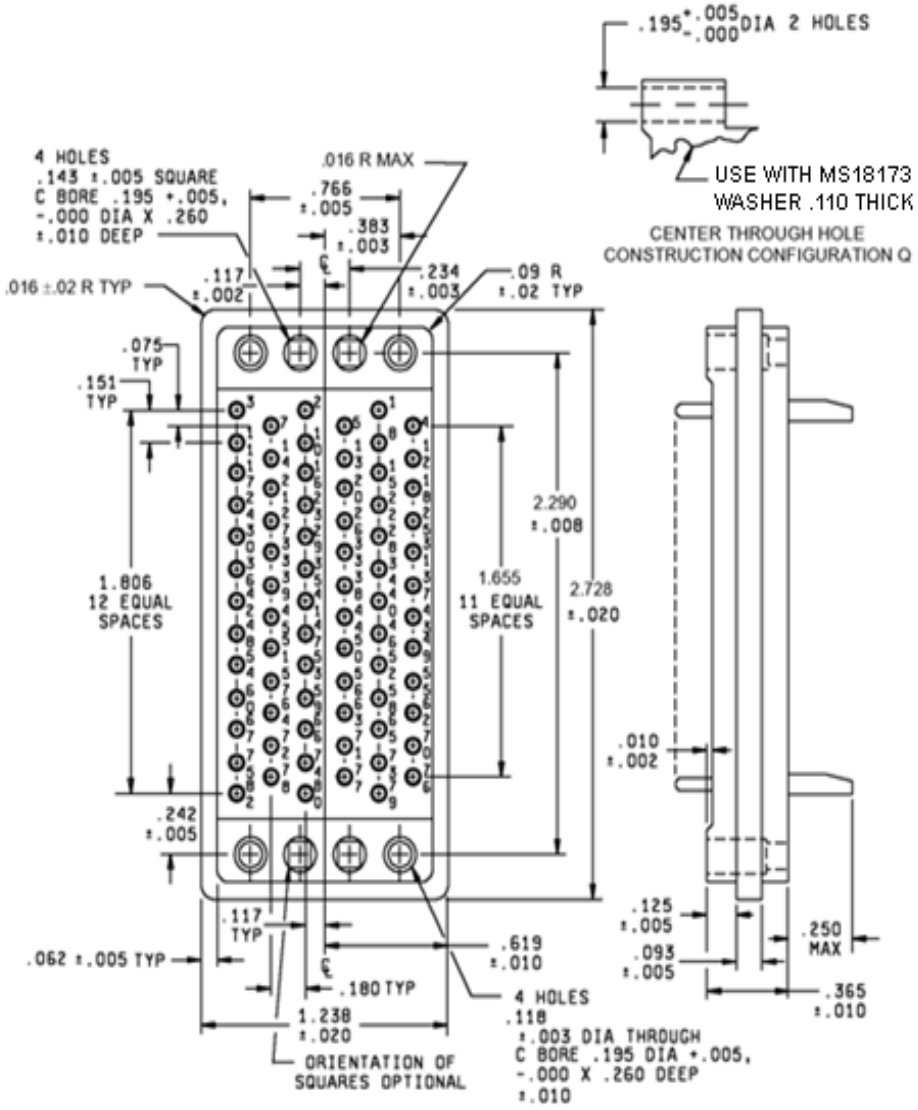
Configuration H - Suitable for all applications, except one-piece jackscrews.
Configuration P - For long jackscrews and shield only (through hole alternate construction).

Insert with 50 contacts

Insert designators H and P

FIGURE 2. Insert arrangements - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:12Z
Check the source to verify that this is the current version before use.



Inches	mm
.002	0.05
.003	0.08
.005	0.13
.008	0.20
.010	0.25
.016	0.41
.02	0.5
.020	0.51
.062	1.57
.075	1.91
.09	2.3
.093	2.36
.110	2.79
.117	2.97
.118	3.00
.125	3.18
.143	3.63
.151	3.84
.180	4.57
.195	4.95
.234	5.94
.242	6.15
.250	6.35
.260	6.60
.365	9.27
.383	9.73
.619	12.72
.766	19.46
1.238	31.45
1.655	42.04
1.806	45.87
2.290	58.17
2.728	69.29

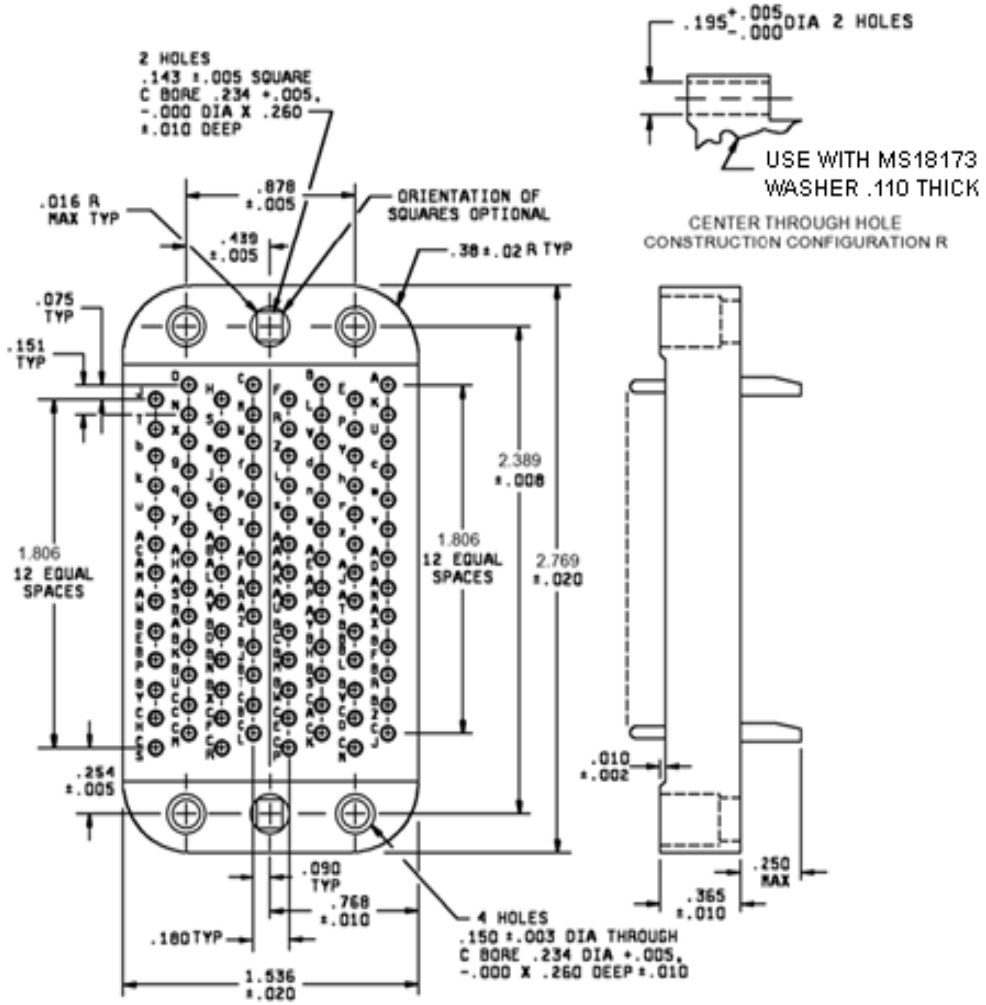
Configuration J - Suitable for all applications, except one-piece jackscrews.
Configuration Q - For long jackscrews and shield only (through hole alternate construction).

Insert with 75 contacts

Insert designators J and Q

FIGURE 2. Insert arrangements - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:12Z
Check the source to verify that this is the current version before use.



Inches	mm
.002	0.05
.003	0.08
.005	0.13
.008	0.20
.010	0.25
.016	0.41
.02	0.5
.020	0.51
.062	1.57
.075	1.91
.090	2.29
.110	2.79
.143	3.63
.150	3.81
.151	3.84
.180	4.57
.195	4.95
.234	5.94
.250	6.35
.254	6.45
.260	6.60
.365	9.27
.38	9.7
.439	11.15
.768	19.51
.878	22.30
1.536	39.01
1.806	45.87
2.389	60.68
2.769	70.33

Configuration K - Suitable for all applications, except one-piece jackscrews.
Configuration R - For long jackscrews and shield only (through hole alternate construction).

Insert with 104 contacts

Insert designators J and Q

FIGURE 2. Insert arrangements - Continued.

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REQUIREMENTS:

Dimensions and configuration: See figures 1 and 2.

Insert material: See MIL-DTL-28748.

Mating connector: See MIL-DTL-28748/4 and MIL-DTL-28748/6.

Part or Identifying Number (PIN): As shown in the following example and table I: 1/

See table I for permissible PINs:

(See PIN code)

M28748/5 -	A	1	H	L	1 A
Basic number of specification sheet	Insert designator	Shield and shield clamp location or retaining plate	Shell and polarization position	Jackscrews or guidepins	Contacts

PIN codes:

Insert designator	Number of contact positions
A	7
B	9
C	14
S	18
D	20
E	26
F	34
G	42
H	50
J	75
K	104
Insert designator through hole 5/	
M	34
N	42
P	50
Q	75
R	104

Shield and shield clamp Location or retaining plate 2/

0 – No shield or retaining plate

Shield and shield clamp location
1 - top MS24132
2 - side MS24132
3 - top MS24133
4 - side MS24133
5 - top MS18192
6 - side MS18192
A - top MS18193
B -side MS18193

Retaining plate
7 - MS18198
8 - MS18199
9 - MS18200

Shell and [polarization designator 3/

0 - No shell included

Plug shell
A - A of MS18203
B - B of MS18203
C - C of MS18203
D - D of MS18203
E - E of MS18203
F - F of MS18203
G - G of MS18203
H - Unpolarized MS18204

Receptacle shell
J - A of MS18204
K - B of MS18204
L - C of MS18204
M - D of MS18204
N - E of MS18204
P - F of MS18204
Q - G of MS18204
R - Unpolarized MS18203

Jackscrews or guidepins 4/

0 – No Jackscrews or guidepins.
F - Fixed jackscrews, MS18196 (not used with shield).
G - Guidepins, MS18197 (not used with shield).
L - Long jackscrews MS18194
S - Short jackscrews MS18195 (not used with shield).

Contacts: In accordance with MIL-DTL-28748.

1A - 100 percent size 20
1L - none included

Jack and guide location: The jackscrew female (turning or fixed) or guide socket shall be located by the A or 1 pin contact.

See footnotes on next page.

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1/ The number zero (0) is used to indicate which parts are not included.
2/ Retaining plate and shells are not available for insert size A.
3/ Polarization positions are for location of plug and receptacle shell pins and slots.
4/ Jackscrew designator L: Multi-piece jackscrews will be supplied with insert designators A through K. Long one-piece jackscrews will be supplied for insert designators M through R.
5/ Inserts M through S have through holes for use with long one-piece jackscrew designs only.
6/ Mold bodies and hardware from different manufacturers are not necessarily interchangeable.

TABLE I. Permissible PIN's (connector assembly).

<table>
<tr><td>Configuration</td><td colspan="5">Accessories</td></tr>
<tr><td>Receptacle</td><td>Insert</td><td>Shield/retaining plate</td><td>Shell</td><td>Jackscrews</td><td>Contacts</td></tr>
<tr><td rowspan="12">M28748/5-</td><td>A</td><td>0, 1, 2</td><td>0</td><td>0, F, G</td><td rowspan="25">1A, 1L,</td></tr>
<tr><td>B</td><td rowspan="2">0, 1, 2, 5, 6, 7</td><td rowspan="12">0, A thru H</td><td rowspan="12"></td></tr>
<tr><td>C</td></tr>
<tr><td>S</td><td>0, 1, 2</td></tr>
<tr><td>D</td><td rowspan="2">0, 1, 2, 5, 6, 7</td></tr>
<tr><td>E</td></tr>
<tr><td>F</td><td rowspan="5">0, 3, 4, 5, 6, 8</td></tr>
<tr><td>G</td></tr>
<tr><td>H</td></tr>
<tr><td>J</td></tr>
<tr><td>L</td><td>0, A, B, 9</td></tr>
<tr><td>Plug</td><td></td><td></td><td></td><td></td></tr>
<tr><td rowspan="13">M28748/5-</td><td>A</td><td>0,1, 2</td><td>0</td><td rowspan="13">0, L, G, S</td></tr>
<tr><td>B</td><td rowspan="2">0, 1, 2, 5, 6, 7</td><td rowspan="12">0, J thru R</td></tr>
<tr><td>C</td></tr>
<tr><td>S</td><td>0, 1, 2</td></tr>
<tr><td>D</td><td rowspan="2">0, 1, 2, 5, 6, 7</td></tr>
<tr><td>E</td></tr>
<tr><td>F</td><td rowspan="4">0, 3, 4, 5, 6, 8</td></tr>
<tr><td>G</td></tr>
<tr><td>H</td></tr>
<tr><td>K</td></tr>
<tr><td>L</td><td>0, A, B, 9</td></tr>
<tr><td>M</td><td rowspan="4">0, 3, 4, 5, 6, 8</td></tr>
<tr><td>N</td></tr>
<tr><td>P</td></tr>
<tr><td>Q</td></tr>
<tr><td>R</td><td>0, A, B, 9</td><td>0</td></tr>
</table>

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For superseded insert Military Standard sheet PIN's to M28748 PIN's see table II.

TABLE II. Supersession data (inserts).

MS PIN	M28748/5 PIN	Number of contact positions
MS24004-1	M28748/5-A0001L	7
MS18275-1	M28748/5-B0001L	9
MS24010-1	M28748/5-C0001L	14
MS24012-1	M28748/5-S0001L	18
MS24014-1	M28748/5-D0001L	20
MS24020-1	M28748/5-E0001L	26
MS24022-1	M28748/5-F0001L	34
MS18258-1	M28748/5-G0001L	42
MS24026-1	M28748/5-H0001L	50
MS18261-1	M28748/5-J0001L	75
MS18263-1	M28748/5-K0001L	104
MS24022-2	M28748/5-M0001L	34 1/
MS18258-2	M28748/5-N0001L	42 1/
MS24052-2	M28748/5-P0001L	50 1/
MS18261-2	M28748/5-Q0001L	75 1/
MS18263-2	M28748/5-R0001L	104 1/

1/ Insert through hole designator. Inserts -2 (M through S) have through holes for use with long one-piece jackscrew designs only.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Reference documents. In addition to MIL-DTL-28748, this document references the following:

MIL-DTL-28748/4
MIL-DTL-28748/6
MS18173
MS18192
MS18193
MS18194
MS18195
MS18196
MS18197
MS18198
MS18199
MS18200
MS18203
MS18204
MS24132
MS24133

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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-2014-037)

Review activities:
Army - AT, MI
Navy - AS, SH
Air Force - 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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